

Rapid Spanning Tree Protocol

- Topics covered:
 - Comparison of STP versions
 - Rapid PVST+
-

Versions of Spanning Tree Protocol

◆ Classic Spanning Tree Protocol (IEEE 802.1D)

This is the original spanning tree protocol.

According to Wikipedia, it was published in 1990, although the original spanning tree was created in 1985, before being standardized.

In classic STP:

- All VLANs share one STP instance

Because there is only one instance:

- We cannot load balance
- We cannot block different ports in each VLAN

◆ Per-VLAN Spanning Tree Plus (PVST+)

Cisco decided to improve upon classic STP.

They developed Per-VLAN Spanning Tree Plus.

Before that, they developed regular PVST, but it only supported ISL, not 802.1Q, so that version is ignored.

PVST+ is Cisco's upgrade to 802.1D.

Each VLAN has its own STP instance.

In previous labs, STP commands included the VLAN number:

```
spanning-tree vlan 1 root primary
```

That is because a separate STP instance runs for each VLAN.

Why This Is Good

- Different ports can be blocked in different VLANs
- Load balancing is possible
- Network bandwidth is used more effectively
- Links are not left unused waiting for failures

STP Speed Limitation

Classic STP and PVST+ are slow.

- Max Age = 20 seconds
- Listening = 15 seconds
- Learning = 15 seconds

It can take up to 50 seconds to react to changes.

This is not fast enough for modern networks.

◆ **Rapid Spanning Tree Protocol (IEEE 802.1w)**

This problem was solved by Rapid STP.

RSTP converges much faster than 802.1D.

However:

- Like 802.1D, it runs one STP instance
- Shared by all VLANs
- Cannot load balance

◆ **Rapid Per-VLAN Spanning Tree Plus (Rapid PVST+)**

Cisco developed Rapid PVST+.

It is Cisco's upgrade to 802.1w.

Features:

- Faster convergence of RSTP
- Separate STP instance per VLAN

Therefore:

- It can load balance, just like PVST+

◆ **Multiple Spanning Tree Protocol (IEEE 802.1s)**

MSTP uses modified RSTP mechanics.

Main improvement:

- Groups multiple VLANs into different instances

Example:

- VLANs 1–5 → Instance 1
- VLANs 6–10 → Instance 2

This allows load balancing.

It is an industry standard and is superior to Rapid PVST+.

If there are many VLANs (for example 200):

- Configuring roots per VLAN is difficult

With MSTP:

- VLANs 1–100 → Instance 1
- VLANs 101–200 → Instance 2

Then configure root bridges per instance.

Cisco devices run the standard 802.1s (no Cisco version).

◆ **Which Version to Use**

- Large networks → MSTP
- Small to medium networks → Rapid PVST+

Rapid PVST+ is also listed in the official exam topics.

◆ Cisco's Summary of RSTP

RSTP is not timer-based like 802.1D.

It improves on the 30+ seconds required by classic STP.

The heart of RSTP is a bridge-to-bridge handshake mechanism.

Ports can move directly to forwarding.

Classic STP:

- Uses long timers
- Slow to ensure no loops

RSTP:

- Actively negotiates
- Moves ports immediately when safe

◆ RSTP and Rapid PVST+

When saying RSTP or Rapid PVST+, it refers to the same behaviour.

Rapid PVST+ operates the same as RSTP, with the addition of:

- A separate instance per VLAN

Industry standards (IEEE)	Cisco versions
Spanning Tree Protocol (802.1D) • The original STP • All VLANs share one STP instance. • Therefore, cannot load balance.	Per-VLAN Spanning Tree Plus (PVST+) • Cisco's upgrade to 802.1D • Each VLAN has its own STP instance. • Can load balance by blocking different ports in each VLAN.
Rapid Spanning Tree Protocol (802.1w) • Much faster at converging/adapating to network changes than 802.1D • All VLANs share one STP instance. • Therefore, cannot load balance.	Rapid Per-VLAN Spanning Tree Plus (Rapid PVST+) • Cisco's upgrade to 802.1w • Each VLAN has its own STP instance. • Can load balance by blocking different ports in each VLAN.
Multiple Spanning Tree Protocol (802.1s) • Uses modified RSTP mechanics. • Can group multiple VLANs into different instances (ie. VLANs 1-5 in instance 1, VLANs 6-10 in instance 2) to perform load balancing.	



◆ Similarities Between STP and RSTP

RSTP:

- Prevents Layer 2 loops
- Elects a root bridge the same way
- Lowest bridge ID becomes root

Root port election:

- Lowest root cost
- Tie-breakers: neighbour bridge ID, then neighbour port ID

Designated port election:

- Lowest root cost
- Tie-breaker: lowest bridge ID

RSTP is an **evolution**, not a revolution.

◆ Port Cost Differences

Classic STP supports speeds up to 10 Gbps.

RSTP expands cost values to support faster speeds.

Speed	STP Cost	RSTP Cost
10 Mbps	100	2,000,000
100 Mbps	19	200,000
1 Gbps	4	20,000
10 Gbps	2	2000
100 Gbps	X	200
1 Tbps	X	20

◆ Port States

Classic STP Port States:

STP Port State	Send/Receive BPDUs	Frame forwarding (regular traffic)	MAC address learning	Stable/ Transitional
Blocking	NO/YES	NO	NO	Stable
Listening	YES/YES	NO	NO	Transitional
Learning	YES/YES	NO	YES	Transitional
Forwarding	YES/YES	YES	YES	Stable
Disabled	NO/NO	NO	NO	Stable

◆ RSTP Port States

RSTP simplifies port states to three.

Blocking and Disabled → Discarding

Listening → Removed

Discarding → no forwarding, no learning

Learning → MAC learning only

Forwarding → forwarding + learning

STP Port State	Send/Receive BPDUs	Frame forwarding (regular traffic)	MAC address learning	Stable/ Transitional
Discarding	NO/YES	NO	NO	Stable
Learning	YES/YES	NO	YES	Transitional
Forwarding	YES/YES	YES	YES	Stable

◆ Port Roles

Original port roles:

- Root
- Designated
- Non-designated

In RSTP:

- Root → unchanged
- Designated → unchanged
- Non-designated → split into:
 - Alternate
 - Backup

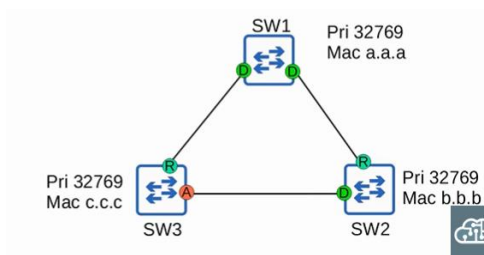
◆ Alternate Port Role

An alternate port:

- Is in discarding state
- Receives a superior BPDUs from another switch

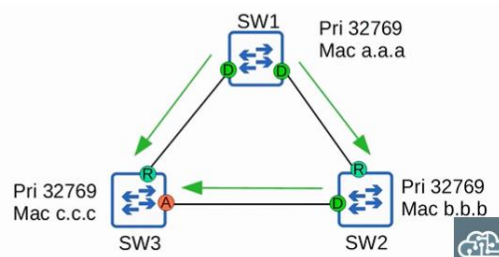
This is the same as a blocking port in classic STP.

Example Topology:



SW3 receives a superior BPDUs from SW2.

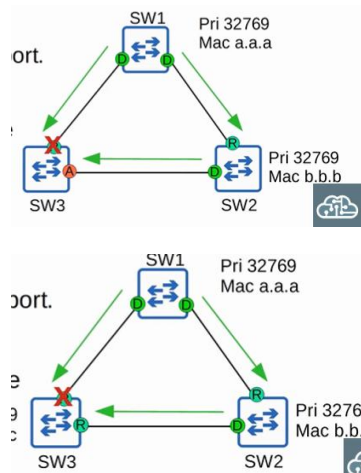
- SW2 interface → Designated
- SW3 interface → Alternate



Alternate ports:

- Act as a backup for the root port
- Move immediately to forwarding if the root port fails

- No transitional states
 - If SW3 root port fails its alternate Root port is ready To immediately become root port without transitional states.



UPLINKFAST:

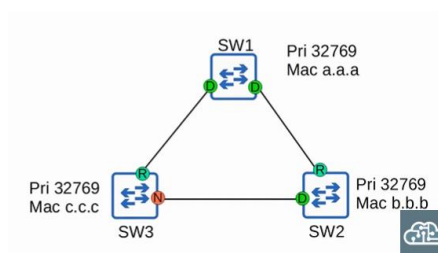
This immediate move to forwarding state functions like a classic STP optional feature called UplinkFast. Because it is built into RSTP, you do not need to activate UplinkFast when using RSTP/Rapid PVST+.

We not look at uplinkfast in the previous, its not mention in CCNA exam topic, So UplinkFast is one STP optional feature that was incorporated into RSTP here is one more that was incorporated into RSTP.

BackboneFast in RSTP:

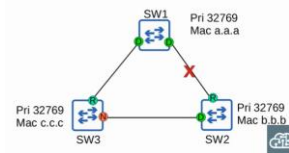
BackboneFast is also built into RSTP.

Understanding with Example:

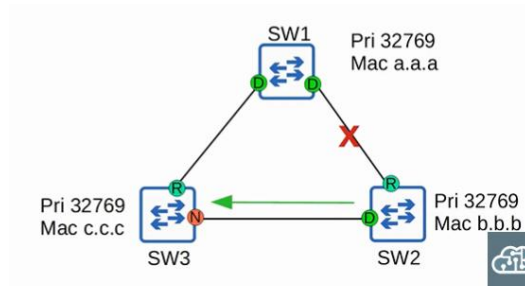


If a root port fails:

- Suppose SW2 root port is cut of so its stop receiving BPDUs from Root Bridge(SW1)



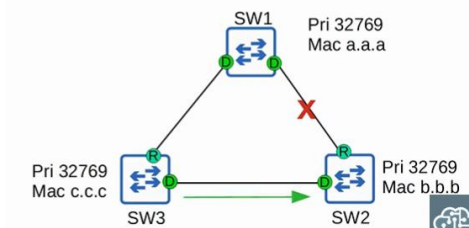
- Then it assumes it is the root, So its sends its own BPDUs to sw3



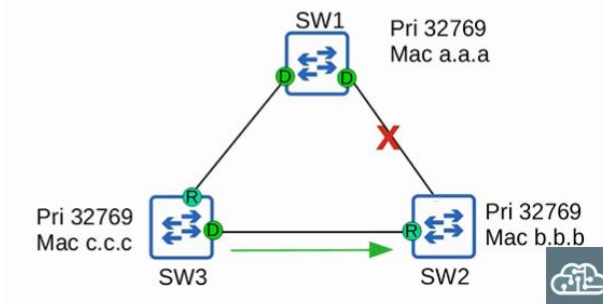
- However, SW3 is now not receiving SW2 BPDUs from both SW1 and SW2, But SW2 BPDUs are inferior they are higher bridge id.

Without BackboneFast:

- SW3 would just ignore these BPDUs from Sw2 until its non-designated port in classic STP
- Finally changes to forwarding state and forward the superior BPDUs to SW2.



- Then SW2 accepts SW1 as its root bridge again



With BackboneFast:

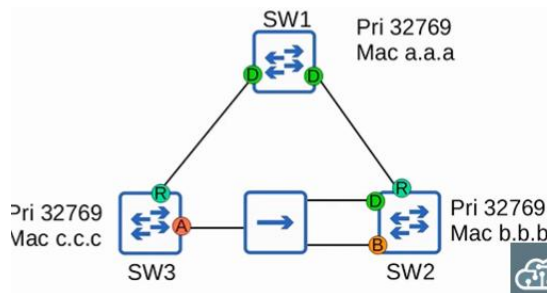
- BackboneFast allow Sw3 to expire the max age timer on that interface and rapidly forward the superior BPDUs to SW2.
- This functionality is built into RSTP, so it does not need to be configured.

◆ Backup Port Role (RSTP)

The RSTP backup port role is a discarding port that receives a superior BPDUs from another interface on the same switch.

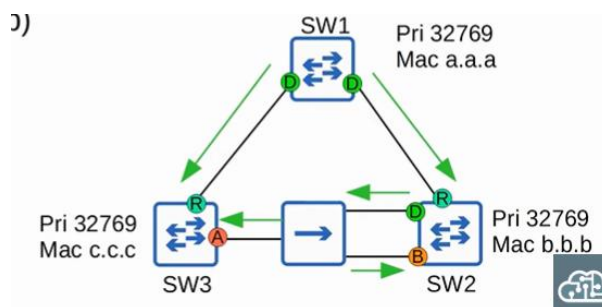
This only happens when two interfaces are connected to the same collision domain, via a hub.

Topology with Hub (Backup Port Scenario)

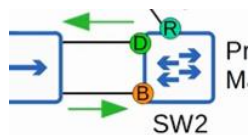


Notice that an Ethernet hub is connected between SW2 and SW3.

When BPDUs are sent in this network:



- The BPDU sent out of SW2's designated port
- Is flooded by the hub
- SW2 receives the same BPDU on another interface



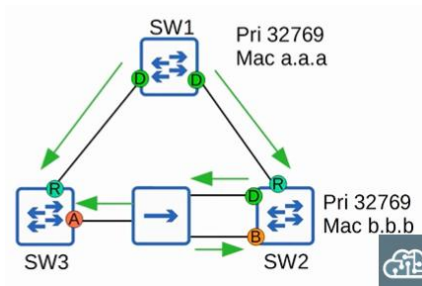
That interface becomes a backup port, not an alternate port.

Important Note About Backup Ports

Hubs are not used in modern networks, so you will probably not encounter a backup port in real networks.

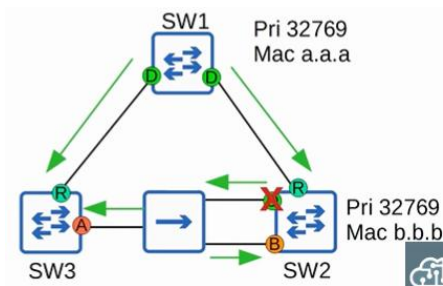
However, it is still something you should know.

Backup Port Behaviour

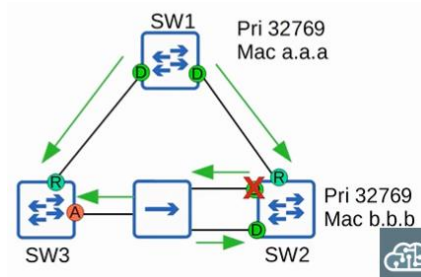


RSTP backup ports function as a backup for a designated port.

If SW2's designated port fails:



- The backup port immediately begins forwarding
- It becomes the designated port



How Designated vs Backup Is Chosen

When two interfaces are connected to the same collision domain:

- The interface with the lowest port ID becomes the designated port
- The other interface becomes the backup port

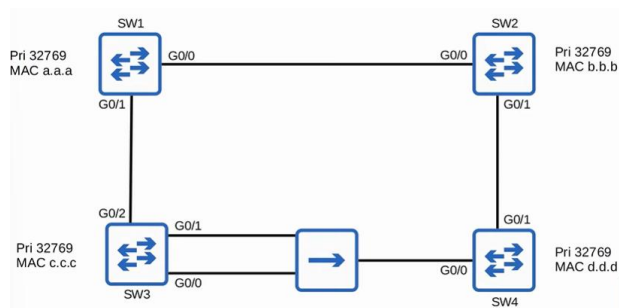
◆ Quiz Question

Identify:

- The **root bridge**
- The **RSTP port role of each switch interface**

Note:

- The hub does not participate in spanning tree
- The hub floods all frames



Answer

Root Bridge Election:

All switches have the same priority.

SW1 has the lowest MAC address, so **SW1 is the root bridge**.

All interfaces on SW1 are designated ports.

Root Ports

SW2 and SW3:

- Root ports are obvious
- They have the lowest root cost

SW4 Root Port Decision

The hub does not add cost.

SW4 chose G0/1 instead of G0/0 because:

- The neighbour bridge ID is lower via G0/1

- SW2 has a lower MAC address than SW3

Therefore:

- SW2's G0/1 → designated
- SW4's G0/1 → root port

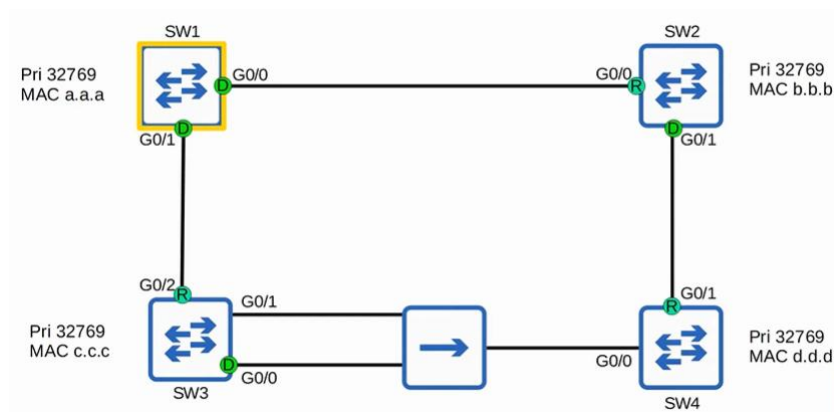
SW3–SW4 Segment

Which switch sets a designated port?

SW3 has a lower root cost, so one of its interfaces becomes designated.

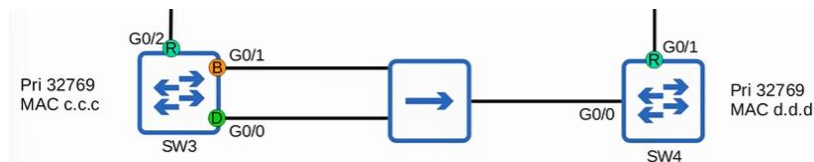
Which one?

- G0/0 has the lower port ID
- G0/0 → designated port



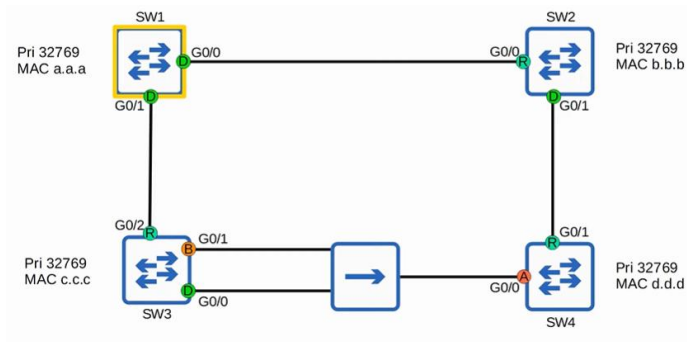
Now:

- SW3 G0/1 receives a superior BPDU from the same switch → backup port



- SW4 G0/0 receives a superior BPDU from a different switch → alternate port





CLI Verification (Rapid PVST+)

On SW3:

```
SW3(config)#spanning-tree mode ?
mst      Multiple spanning tree mode
pvst     Per-Vlan spanning tree mode
rapid-pvst Per-Vlan rapid spanning tree mode
SW3(config)#spanning-tree mode rapid-pvst
```

Rapid-PVST is the default, so you have to not run this command, but we do:

“spanning-tree mode rapid-pvst”

Then: “show spanning-tree”

Output shows:

```
SW3(config)#do show spanning-tree

VLAN0001
Spanning tree enabled protocol rstp
  Root ID    Priority    32769
             Address     aaaa.aaaa.aaaa
             Cost        4
             Port        3 (GigabitEthernet0/2)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
             Address     cccc.cccc.cccc
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time   300 sec

Interface     Role Sts Cost      Prio.Nbr Type
-----
Gi0/0         Desg FWD 4         128.1   Shr
Gi0/1         Back BLK 4         128.2   Shr
Gi0/2         Root FWD 4         128.3   P2p
```

##Spanning tree enabled protocol rstp##

Although it says rstp, this is Cisco Rapid PVST+.

Backup Port in CLI Output

SW3 G0/1:

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Desg	FWD	4	128.1	Shr
Gi0/1	Back	BLK	4	128.2	Shr
Gi0/2	Root	FWD	4	128.3	P2p

- Role: Backup
- Status shown as BLK

Note:

- BLK means discarding in RSTP terminology

Alternate Port in CLI Output

On SW4: I run command “show spanning-tree” command also.

```
SW4#show spanning-tree

VLAN0001
Spanning tree enabled protocol rstp
Root ID    Priority    32769
           Address    aaaa.aaaa.aaaa
           Cost      8
           Port      2 (GigabitEthernet0/1)
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
           Address    dddd.ddd.ddd
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time 300 sec

Interface   Role Sts Cost    Prio.Nbr Type
-----
Gi0/0       Altn BLK 4       128.1   P2p
Gi0/1       Root FWD 4       128.2   P2p
```

- G0/0 is an Alternate port
- Status shown as BLK
- BLK = discarding in RSTP

Compatibility with Classic STP

Rapid STP is compatible with Classic STP. The interface(s) on the Rapid STP-enabled switch connected to the Classic STP-enabled switch will operate in Classic STP mode (timers, blocking → listening → learning → forwarding process, etc).



Rapid STP is compatible with classic STP.

If a rapid STP switch connects to a classic STP switch:

- That interface runs in classic STP mode
- Uses classic timers and states

Other interfaces can still run rapid STP.

RSTP BPDUs vs Classic STP BPDUs

◆ BPDUs Comparison of RSTP & Classic STP

Most BPDUs fields remain unchanged.

<pre>> Frame 999: 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface 0 > Ethernet II, Src: aa:aa:aa:aa:aa:ab (aa:aa:aa:aa:aa:ab), Dst: PVST+ > 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 10 > Logical-Link Control > Spanning Tree Protocol Protocol Identifier: Spanning Tree Protocol (0x0000) Protocol Version Identifier: Spanning Tree (0) BPDUs Type: Configuration (0x00) BPDUs flags: 0x00 0... .. = Topology Change Acknowledgment: No = Topology Change: No Root Identifier: 32768 / 10 / aa:aa:aa:aa:aa:aa Root Bridge Priority: 32768 Root Bridge System ID Extension: 10 Root Bridge System ID: aa:aa:aa:aa:aa:aa (aa:aa:aa:aa:aa:aa) Root Path Cost: 0 Bridge Identifier: 32768 / 10 / aa:aa:aa:aa:aa:aa Bridge Priority: 32768 Bridge System ID Extension: 10 Bridge System ID: aa:aa:aa:aa:aa:aa (aa:aa:aa:aa:aa:aa) Port Identifier: 0x8002 Message Age: 0 Max Age: 20 Hello Time: 2 Forward Delay: 15</pre>	<pre>> Frame 71: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0 > IEEE 802.3 Ethernet > Logical-Link Control Spanning Tree Protocol Protocol Identifier: Spanning Tree Protocol (0x0000) Protocol Version Identifier: Rapid Spanning Tree (2) BPDUs Type: Rapid/Multiple Spanning Tree (0x02) BPDUs flags: 0x3c, Forwarding, Learning, Port Role: Designated 0... .. = Topology Change Acknowledgment: No .0... .. = Agreement: No ..1... .. = Forwarding: Yes ...1... .. = Learning: Yes 11... .. = Port Role: Designated (3) 0... .. = Proposal: No 0... .. = Topology Change: No Root Identifier: 32768 / 1 / aa:aa:aa:aa:aa:aa Root Bridge Priority: 32768 Root Bridge System ID Extension: 1 Root Bridge System ID: aa:aa:aa:aa:aa:aa (aa:aa:aa:aa:aa:aa) Root Path Cost: 4 Bridge Identifier: 32768 / 1 / cc:cc:cc:cc:cc:cc Bridge Priority: 32768 Bridge System ID Extension: 1 Bridge System ID: Silicon_cc:cc:cc:cc:cc:cc (cc:cc:cc:cc:cc:cc) Port identifier: 0x8001 Message Age: 1 Max Age: 20 Hello Time: 2 Forward Delay: 15 Version 1 Length: 0</pre>
---	--



Key differences to know:

Protocol Version

- Classic STP → Version 0
- Rapid STP → Version 2

(Remember for exam)

BPDUs Type

- Rapid STP BPDUs type = 2

BPDUs Flags

- Classic STP uses 2 bits
- Rapid STP uses all 8 bits

These flags are used for negotiation, allowing fast convergence.

BPDU Origination

Classic STP:

- Only the root bridge originates BPDUs

Rapid STP:

- All switches originate and send BPDUs from designated ports

♦ Faster Aging in RSTP

Classic STP:

- Neighbour considered lost after 20 seconds

Rapid STP:

- Neighbour considered lost after 3 missed BPDUs (6 seconds)

When neighbour is lost:

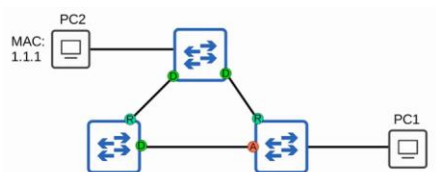
- Switch flushes MAC addresses learned on that interface

Reason:

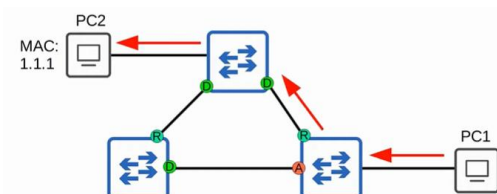
- It can no longer reach anything through that interface

Topology Change Example

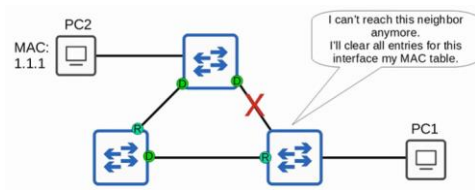
Normal traffic path:



PC1 → SW → SW → PC2



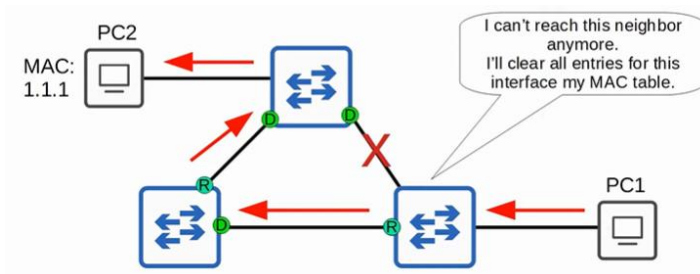
If a link fails:



- Switch detects neighbour loss
- Clears MAC entries
- Another interface becomes root port

Traffic is flooded again until MACs are relearned:

PC1 → new path → PC2



◆ Final Notes

- RSTP handles topology changes much faster than classic STP
- Deep details are not required for CCNA
- CCNP / CCIE requires deeper study

RSTP – Link Types

◆ Types of Links in RSTP STP

RSTP distinguishes between three different link types.

1. Edge Link Type
2. Point-to-Point Link Type
3. Shared Link Type

◆ Edge Link Type

An edge port is a port that is connected to an end host.

It moves directly to forwarding without negotiation.

This should sound familiar.

It sounds like PortFast.

PortFast functionality was built into RSTP.

So now three classic STP optional features are built into RSTP by default:

- UplinkFast
- BackboneFast
- PortFast

Edge Port Behaviour:

Because there is no risk of a loop, edge ports:

- Skip negotiation
- Move straight to forwarding

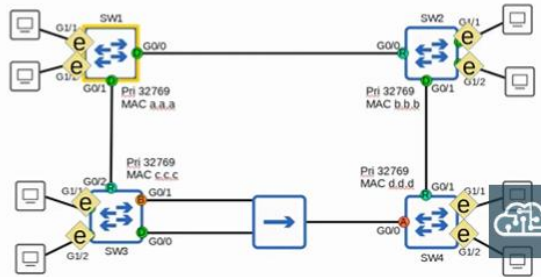
They function like a classic STP port with PortFast enabled.

In fact, you configure an edge port simply by enabling PortFast, just like in classic STP.

```
SW1(config-if)# spanning-tree portfast
```

So PortFast and an RSTP edge port are the same thing.

Edge Port Topology:



In this network, all ports connected to PCs should be configured as edge ports.

◆ Point-to-Point Link Type

Point-to-point links are used for direct connections between two switches.

Because the connection is to another switch (not a hub):

- They operate in full-duplex

You do not need to configure the interface as point-to-point.

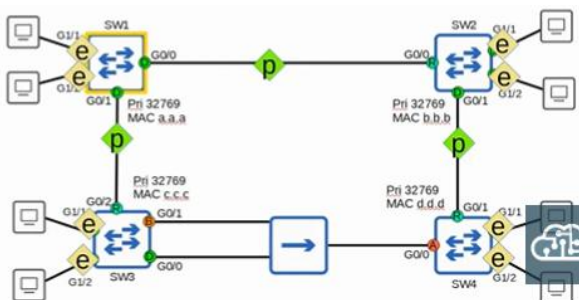
The switch can detect that it is connected to another switch and will operate as a point-to-point link automatically.

Manual Configuration (Optional):

If you want to explicitly configure it:

```
SW1(config-if)# spanning-tree link-type point-to-point
```

Point-to-Point Links in the Topology



These direct switch-to-switch connections are point-to-point links.

◆ Shared Link Type

Shared links are connections to a hub.

Because hubs cause collisions:

- These links must operate in half-duplex

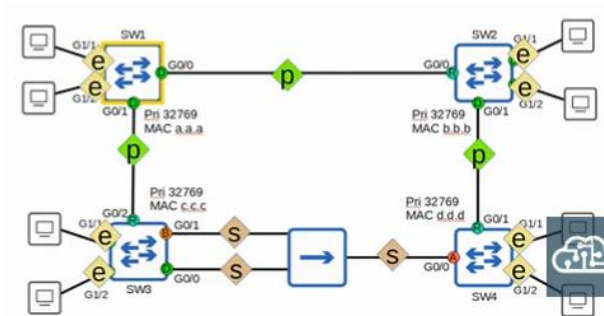
Once again, you do not need to configure this manually.

The switch will detect it automatically.

Manual Configuration (Optional):

```
SW1(config-if)# spanning-tree link-type shared
```

Shared Link Topology:



All connections attached to the hub are shared links.

Although you should know about this link type, you will probably never see it in real networks, because hubs are no longer used.

◆ Summary of These Topics:

First, we compared different versions of STP.

- Classic STP → IEEE 802.1D
- Cisco upgrade → PVST+ (per-VLAN instance)

Next:

- Rapid STP → IEEE 802.1w
- Cisco version → Rapid PVST+ (per-VLAN instance)

Then:

- MSTP → IEEE 802.1s
- Allows multiple instances and VLAN grouping
- Cisco runs the industry standard MSTP

Rapid STP Overview

All information applies to RSTP and Rapid PVST+.

RSTP is an evolution, not a revolution.

Instead of timers, it uses a negotiation process to:

- Move ports quickly to forwarding
- React rapidly to topology changes

Detailed negotiation is not required for CCNA.

RSTP Port States

RSTP has three port states:

- Discarding
- Learning
- Forwarding

The listening state was removed.

Learning is often skipped due to built-in features like:

- UplinkFast
- BackboneFast

RSTP Port Roles

There are four port roles:

- Root
- Designated
- Alternate
- Backup

Root and designated roles are unchanged.

Alternate ports:

- Receive a superior BPDU from another switch

Backup ports:

- Receive a superior BPDU from the same switch
- Only occur when connected to a hub

Built-In STP Features in RSTP

Classic STP optional features now built into RSTP:

- UplinkFast
- BackboneFast
- PortFast (via edge ports)

You must know PortFast for CCNA.

Detailed knowledge of UplinkFast and BackboneFast is not required.

RSTP BPDU Differences

RSTP BPDU:

- Protocol version = 2

Classic STP BPDU:

- Protocol version = 0

Remember these numbers for the exam.

Important point:

- In RSTP, all switches send BPDUs
- In classic STP, only the root bridge sends BPDUs

RSTP Link Type

- Edge → connected to end hosts, configured using PortFast
- Point-to-Point → switch-to-switch, full-duplex
- Shared → connected to hubs, half-duplex

Hubs are obsolete, so shared links are rarely seen.